

Patterning of Quantum Dots Light-emitting Diodes Based on IGZO Films

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Abstract

Patterned quantum dots light-emitting diodes (QLED) with sputtered IGZO films as electron transport layers are fabricated. The patterned IGZO films can define pixels ~10 μm, while accompanied by the nonuniform current spreading effect. The localized current density distribution observed by luminance distribution during the aging degrades the device seriously. In order to overcome this issue, the patterned SiO₂ layer as the bank is introduced. This work also shows a potential way to tune the out couple efficiency and micro-cavity effect (optical length) in QLED by charge transport layer thickness.

Author Keywords

Quantum dots light-emitting diodes; Patterning; IGZO; Sputter;

1. Introduction

Quantum dots light-emitting diodes (QLED) have attracted much attention of researchers and industrial insiders because of the unique opt-electric properties, including narrower emission, lower turn-on voltage, and low-cost solution process, etc.(1) Despite tremendous efforts have been put into this field and the device performance has already been promoted a lot, there are still several obstacles hindering the QLED application.(2) Among them, luminance stability and pixel patterning are essential for the display application.(3,4) Generally, the performance and stability of QLED are strongly affected by electron transport layers (ETL) usually are metal oxide nanocrystals (NCs) including ZnO NCs and ZnMgO NCs.(5) There are a number of reports focusing on stability enhancement by optimizing ETL since the active chemical properties of ZnO-based nanocrystals and the sensitivity to the environment (temperature, oxygen, moisture, etc.).(6–10) However, it is difficult to define pixels by spin-coating ETL layer.(11–14) The patterned QLED based on patterned ETL with good performance and stability is rarely reported.(15,16) On the other hand, the all solution patterning technique like ink-jet printing, photolithography, etc. is still in development.(17–20) This makes the stability and patterning hard to be compatible in current QLED fabrication.

In this work, we report a patterned QLED with industrialized materials, indium gallium zinc oxide (IGZO) films, as ETL. The pixel patterning process is fully compatible with the general oxide thin-film transistor (TFT) process and can easily realize high pixel density with size ~10 μm under our current process condition. The whole patterns are defined by IGZO film

thickness, which is implemented through sputtering and photolithography. The detailed aging behaviors also characterized by luminance distribution. The luminance degradation mechanisms for such QLED with different patterned ETL are discussed. This work also shows a potential way to tune the out couple efficiency and micro-cavity effect (optical length) in QLED using ETL thickness.

2. Experimental

The inverted QLED are fabricated on ITO glass with pre-patterned IGZO films. The experimentation is described in details.

Films deposition and characterization: The IGZO films are deposited by radio frequency (RF) magnetron sputtering (Kurt J. Lesker PVD 75 pro line) at room temperature in pure Ar atmosphere (5 mTorr) and the thickness is controlled by deposition time with constant power (40 W). In order to get films with high quality, the work distance and deposition angle is tuned to obtain lower deposition rate about 1.5 nm/min. All films' thickness is checked by the step profiler (KLA-Tencor P-6). After deposition, the film is characterized by UV-Vis spectrometer, atomic force microscopy (AFM, Asylum Research, MFP-3D Stand Alone), Hall effect measurements, etc.

Substrate patterning: The patterned ITO glass (10 Ω/□) are prepared by laser ablation to define the cathode. The first layer, 60 nm IGZO film, is deposited on the ITO substrate. Then the second layer 30 nm IGZO film is patterned by photolithography, sputter and lift-off. In the process, the thick photoresist, 4 μm, is coated due to facilitate the following lift-off process. The higher energy plasma during the sputter will damage the photoresist and cause difficulties in lift-off, although the lower IGZO film deposition rate is tuned here. In this way, ~10 μm pattern with good uniformity is easily fabricated. During the process, the positive RZJ-304 photoresist exposed by Mask Aligner Karl SUSS MA/BA 6 and developed by RZX-3038 (TMAH, 2.38%). After the substrates prepared, they are dicing by Optium ADS-800 to meet the device fabrication process.

Device fabrication and characterization: The QLED with inverted architecture are fabricated on the patterned substrate. The whole device structure is ITO/IGZO (60 nm, 90 nm)/QDs/TcTa (10 nm)/NPB (20 nm)/HAT-CN(20 nm)/Al, in which organic hole transport layers (HTL), TcTa/NPB/HAT-CN, are evaporated and the IGZO films are sputtered. The devices are package by UV resin (LOCTITE 3335), which has little impact on the device performance. In this work, the quantum dots are supplied by Shenzhen Planck Innovation

Technologies Co. Ltd. The HTL are purchased in Luminescence Technology Corp. After fabrication, the devices basic performance is characterized by a calibrated QLED integrating sphere test system (Ocean Optics). The luminance distribution is characterized by an imaging luminance meter.

3. Results and Discussion

The quality of substrate patterns is essential for device performance. As shown is Figure 1(a), the process defines pattern by photolithography and lift-off, which is a standard process in the microelectronics industry. In fact, limited by our condition, this patterning process is more suitable to be completed by etching. Owing to the mature TFT process, the pattern can be formed with better quality and smaller size. Figure 1(b) shows the image of the patterned substrate. The patterns are with good uniformity (dark spots are dust on objective lens). This means the thicker photoresist and lower deposition rate benefit the lift-off, although in our limited process condition. It's worth noticing that the edge and corner of the patterns are not smooth enough and also some impurities can be seen. These defects will also influence the device's performance and stability, which can be verified by luminance distribution results in Figure 2(b).

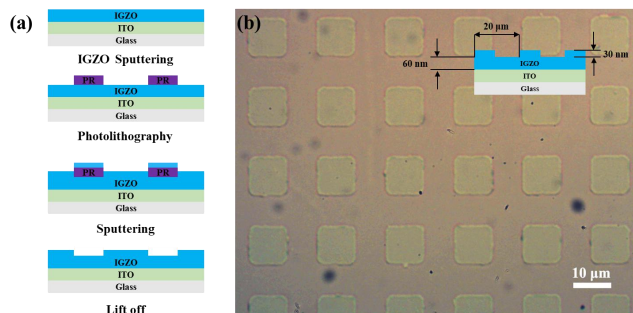


Figure 1. (a) Process flow of ETL (IGZO films) patterning, (b) Optical Microscope image of patterned substrate.

Since the excellent electron transport properties, the IGZO films is more suitable for TFT application. In QLED, the carriers are usually unbalanced injection. The excess electron will cause serious nonradioactive (Auger) recombination. The distinguishing charge transport capacity between IGZO films ($\mu_e \sim 10 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$) and organic hole transport layers ($\mu_h \sim 10^{-4} \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$) worsens the situation.(21–24)

Figure 2 demonstrates the basic performance of IGZO based patterned QLED. The whole device with IGZO pattern (Figure 2 (a)) shows a patterned emission due to the thickness variation in Figure 2 (b). The area with thinner IGZO shows stronger emission and the thicker area shows weaker. This difference is caused by the current distribution, since area with thicker IGZO is with higher resistance along the carrier transport path, according to the Ohm's law ($R = \rho L/S$). In this way, the fine patterns can be displayed.

Figure 2(c) and (d) show the J-V-L and external quantum efficiency (EQE) of the device with and without patterns. The EQE is 2.0% for the device with the patterns (IGZO Array), which is two-thirds comparing with the other. The degradation of the efficiency compared with the normal devices is attributed to the current crowding effect and efficiency roll-off at high current density.(25–28) The leakage originates from structure variation also plays a part. The localized current density will also aggravate the Auger recombination. Those negative effects

should be aware of in future scaling up of high PPI (pixels per inch) QLED. It is also the possible reason for the short lifetime of ink-jet printed QLED. The uniform current spread is necessary for high reliability device, while the structure variation is hard to control during the device fabrication. In this work, we use the IGZO film as ETL and the evaporated organic hole transport layer, which is easy form uniform layer can benefit the uniform current spread. The current control is completed by ETL thickness.

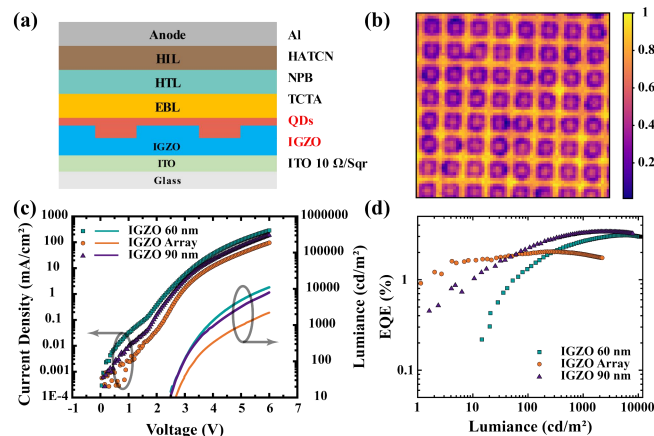


Figure 2. (a) Device architecture, (b) the luminance distribution of pixel arrays (pixel size $\sim 10 \mu\text{m}$), (c) J-V-L and (d) EQE of the device.

In order to investigate the localized current density effect and luminance stability of the device, continuous luminance distribution monitoring is performed. The degradation before and after 8 hours' constant voltage stress at 6V is observed, as shown in Figure 3(a) and Figure 3(b). As shown, the thinner area shows higher luminance than the thicker area (pixels), while after the aging test, the thinner layer shows lower luminance. The entire luminance is degraded $\sim 50\%$. Noting that the luminance data is normalized by the max luminance spot before the aging. The radial patterns are caused by impurities on the substrate during the patterning process, which affects the QDs' spin coating. This "volcanic" type luminance distribution, dark spot center, has also been observed in organic light-emitting diodes (OLED).(29) The impurities on the substrate will cause current crowding and accumulation of joule heat. This will degrade the device stability rapidly. So the patterning process should better be carried out in a strict cleaning room and using etching rather than lift-off. Figure 3(c) shows the detailed luminance changing process surround the dark spot before and after the aging test. It can be seen that after 8h aging the area with higher initial luminance change to lower and the lower initial luminance area become higher. This is in accordance with the J-V-L shown in Figure 1. Since the initial higher luminance area is with higher current density (Ohm's law), the degradation rate will also be higher than an area with the initial lowering luminance area. So, after aging, the emission situation is reversed. The localized current density effect causes serious degradation and should be avoided in all patterned devices not only through patterning charge transport layers but also electrode or by ink-jet printing. From this perspective, the thickness control of the charge transport layer is more suitable for line-width narrower by tuning the micro-cavity effect (optical path) similar to OLED, while combine with the ink-jet printing technique.(30)

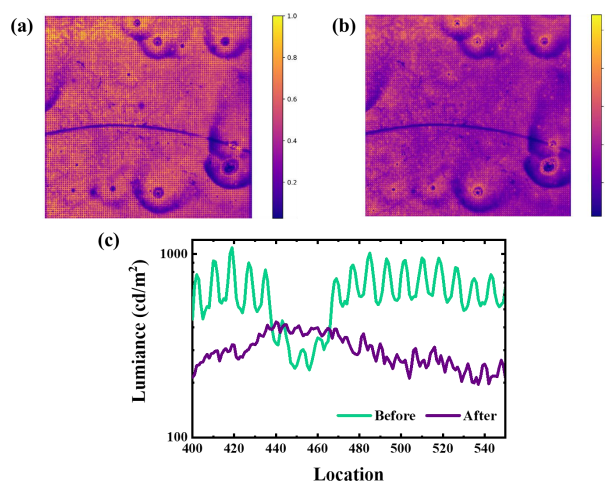


Figure 3. (a) Patterned device luminance distribution before aging, (b) After 8 hours' constant voltage aging at 6V. (c) Luminance distribution details comparison before and after aging test.

On the other hand, the 10μm size patterns are defined, while the leakage and contrast ratio is limited. The reason is the thinner ETL area is also emitting light. To solve this problem, we use SiO₂ layer as a bank and following the same patterning process. Since the SiO₂ or other dielectric layers are often used in the TFT process to form gate isolation. This process also is compatible with the general TFT process. Figure 4 shows the image of QLED pixels (80 μm x 180 μm) fabricated in this way and the luminance of the device. The quality is much more improved compared with the device without SiO₂ bank (Figure 2(b)), since the bank can efficiently suppress leakage and enhance contrast ratio.

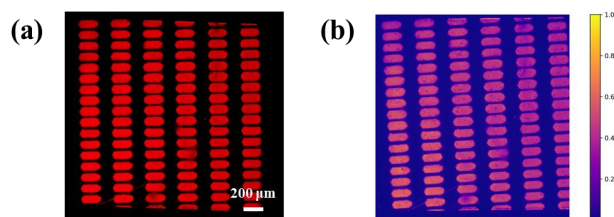


Figure 4. (a) Optical image of device with IGZO ETL and SiO₂ as a bank (Pixel 80 μm x 180 μm, Active Area: 2 mm x 2 mm). (b) Normalized luminance distribution of the device. The pixels are much more improved compared with device without SiO₂ bank.

4. Conclusion

In summary, the inverted QLED with sputtered IGZO films as ETL are fabricated. The basic performance of the devices and patterning methods which are compatible current TFT process is shown. The localized current density effects and relation between luminance distribution and current distribution are shown. The aging behavior that initial higher luminance changing to lower after aging is due to the nonuniform stress. Using SiO₂ as bank can suppress the current nonuniform spread and leakage. The strategy controlling of charge transport layer thickness also shows a way to tune micro-cavity effect (optical length) in QLED.

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